

Middle Convolution of ℓ -adic Perverse Sheaves

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1 Introduction

1.1 Notations

1.1.1. Motivated by the growing interest in the formalization of mathematics within the mathematical community, we adopt a style of notation inspired by type theory throughout this write-up, without committing to type-theoretic foundations. The symbols below denote types and/or categories depending on your philosophical inclination, where $X : T$ denotes that X is a term of type T , or an object of the category T .

Set	Sets
Set^{fin}	Finite sets
Top	Topological spaces
Grp	Groups
Grp^{top}	Topological groups
Grp^{pro}	Profinite groups
Grp^{ab}	Abelian groups
Ring	Rings (associative, unital, not necessarily commutative)
Ring^{top}	Topological rings
Field	Fields
$\text{Field}^{\text{top}}$	Topological fields
Mod_R	Left modules over $R : \text{Ring}$
Vect_k	Vector spaces over $k : \text{Field}$
$\text{Vect}_k^{\text{fin}}$	Finite-dimensional vector spaces over $k : \text{Field}$
$\text{Rep}_\Lambda(G)$	Continuous Λ -linear representations of $G : \text{Grp}^{\text{top}}$, where $\Lambda : \text{Ring}^{\text{top}}$
$\text{Rep}_\Lambda^{\text{fin}}(G)$	Continuous Λ -linear representations of $G : \text{Grp}^{\text{top}}$, where $\Lambda : \text{Ring}^{\text{top}}$, whose underlying topological Λ -modules are finitely generated
Sch	Schemes
Sch_S	Schemes over base scheme $S : \text{Sch}$
Var_k	Varieties over $k : \text{Field}$, i.e. integral separated schemes of finite type over k
Curv_k	Curves over $k : \text{Field}$, i.e. one-dimensional varieties over k
$\text{Curv}_k^{\text{sm}}$	Smooth curves over $k : \text{Field}$
$\text{Man}_{\mathbb{C}}$	Complex manifolds
Riem	Riemann surfaces, i.e. connected one-dimensional complex manifolds
Site	Sites, i.e. categories equipped with Grothendieck topologies
$\text{Sh}(X, A)$	Sheaves on $X : \text{Site}$ with values in a category A admitting small limits
$\text{Mod}(X, \mathcal{R})$	Sheaves of left $\mathcal{R}$ -modules on $X : \text{Site}$, where $\mathcal{R} : \text{Sh}(X, \text{Ring})$
$D^b(A)$	Bounded derived objects of an abelian category A
$D^b(X, k)$	Bounded derived objects of $\text{Sh}(X, \text{Vect}_k)$, where $X : \text{Site}$ and $k : \text{Field}$
$X \rightarrow Y$	Morphisms from $X : T$ to $Y : T$

1.1.2 Global Notations. Let $X, Y : \text{Site}$, $k : \text{Field}$, $n : \mathbb{Z}$, $V : \text{Vect}_k$, and $f : X \rightarrow Y$. Also let $T : \text{Top}$ with base point $t : T$. The following notations will be used throughout the write-up.

$(-)^{\text{an}} : \text{Curv}_{\mathbb{C}}^{\text{sm}} \rightarrow \text{Riem}$	The complex analytification functor
$\pi_1^{\text{top}}(T, t) : \text{Grp}^{\text{top}}$	The topological fundamental group, equipped with the discrete topology
$[n] : \text{D}^b(X, k) \rightarrow \text{D}^b(X, k)$	The cohomological shift functor by n
$\Gamma(X, -) : \text{D}^b(X, k) \rightarrow \text{D}^b(\text{Vect}_k)$	The derived global sections functor
$\mathcal{H}^n : \text{D}^b(X, k) \rightarrow \text{Sh}(X, \text{Vect}_k)$	The n -th cohomology sheaf functor
$\text{H}^n(X, -) : \text{D}^b(X, k) \rightarrow \text{Vect}_k$	The n -th hypercohomology functor
$\underline{V} : \text{Sh}(X, \text{Vect}_k)$	The constant sheaf associated with V
$f_* : \text{D}^b(X, k) \rightarrow \text{D}^b(Y, k)$	The derived pushforward along f
$f^* : \text{D}^b(Y, k) \rightarrow \text{D}^b(X, k)$	The derived pullback along f
$\otimes_k : \text{D}^b(X, k) \times \text{D}^b(X, k) \rightarrow \text{D}^b(X, k)$	The derived tensor product
$\mathcal{H}om : \text{D}^b(X, k)^{\text{op}} \times \text{D}^b(X, k) \rightarrow \text{D}^b(X, k)$	The derived sheaf Hom
$\mathcal{E}nd : \text{D}^b(X, k) \rightarrow \text{D}^b(X, k)$	The derived sheaf End

where the derived functors are assumed to preserve boundedness.

1.2 Motivations

1.2.1. Monodromy groups are interesting mathematical objects that arise naturally in both geometric and arithmetic contexts. Roughly speaking, they describe how locally defined data change when transported along a loop. Controlling and understanding monodromy groups has been a central theme in both of these worlds, with significant applications. We will discuss both the geometric and arithmetic sides, but this write-up will primarily focus on the arithmetic side, with the geometric side serving as motivations and intuitions.

1.2.2. Let $w : \mathbb{C}$ be nonzero, and consider the equation

$$z^2 = w$$

1.2.3 A Question in Arithmetic Geometry. Here is a

1.3 Structure of the Write-up

1.3.1. In § 2

1.3.2. In § 3

1.3.3. In § 4

1.3.4. In § 5

1.3.5. In § 6

1.4 Acknowledgements

1.4.1. The author thanks his advisor, Prof. Chris Hall, for his guidance and advice.

1.4.2 **Typesetting.** The typesetting style of this write-up is inspired by [Kat88].

1.4.3 **Declaration of AI Use.** The use of generative AI in the preparation of this write-up was limited to the following purposes: polishing wording, phrasing, spelling, grammar, and formatting; offering writing advices; answering mathematical questions from the author to help him learn the subject; searching and citing appropriate references; translating reading materials in French into English; retypesetting old references with archaic typesettings for ease of reading; providing emotional support for the author; providing fitness advice for the author to stay healthy and energetic while working on the write-up.

2 Local Systems and Lisse Sheaves

2.1 Local Systems in Topology

2.1.1 **Notation.** Define the following collection of notations

$X : \text{Curv}_{\mathbb{C}}^{\text{sm}}$	A smooth projective connected curve over $\mathbb{C}$
$S : \text{Set}^{\text{fin}}$	A finite nonempty subset of closed points of X
$U : \text{Curv}_{\mathbb{C}}^{\text{sm}}$	The open subvariety $X \setminus S$
$j : U \rightarrow X$	The inclusion map
$m : \mathbb{N}$	The cardinality of S

2.2 Local Systems in the ℓ -adic Setting

2.3 Local Systems in

3 Perverse Sheaves and the Six Functor Formalism

4 Middle Convolution and Monodromy Groups

5 Mod- ℓ Monodromy of Hyperelliptic Curves

6 Formalization in Lean

References

[Kat88] Nicholas M. Katz. *Gauss Sums, Kloosterman Sums, and Monodromy Groups*, volume 116 of *Annals of Mathematics Studies*. Princeton University Press, 1988.